

Using **GitHub Projects** to track defect creation, bug fixes, and new feature development is a great way to organize your software development lifecycle. Here's a structured way to do it:

Step-by-Step Setup Guide

1. Set Up a GitHub Project Board

- Go to your repository or organization → **Projects** tab → Click “**New project**”
 - Choose “**Table**” or “**Board**” view (Board is Kanban-style, good for task tracking)
 - Name your project (e.g., *Product Roadmap*, *Sprint 2025*, *Bug Tracker*)
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2. Define Columns or Statuses

For a **Kanban-style board**, define columns like:

- Backlog
- To Do
- In Progress
- In Review
- Done

You can also use more specific columns if needed:

- Defects / Bugs
- New Features

- QA Testing

3. Use GitHub Issues for Work Items

Create issues to track individual work items. Label them clearly:

- `bug` for defects
- `enhancement` or `feature` for new features
- `documentation`, `testing`, etc.

Use templates for consistency:

- `bug_report.md`
- `feature_request.md`

Each issue should include:

- Title
- Description
- Steps to reproduce (for bugs)
- Acceptance criteria

4. Use Labels to Categorize

Create and apply consistent labels:

- `type: bug`, `type: feature`, `type: tech-debt`

- `priority: high, priority: low`
 - `status: triaged, status: blocked`
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5. Automate Workflows (Optional but Powerful)

Use **GitHub Actions** and **Project Automation** to:

- Automatically move issues between columns
- Close issues when PR is merged
- Assign reviewers

Example:

```
on:
  pull_request:
    types: [closed]
jobs:
  close-linked-issues:
    runs-on: ubuntu-latest
    steps:
      - name: Close linked issues
        uses: peter-evans/close-issue@v1
```

6. Link Issues to Pull Requests

When fixing bugs or implementing features:

Reference the issue in the PR description:

Fixes #123

Closes #456

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This links and auto-closes issues upon merge.

7. Use Milestones (Optional)

Group issues and PRs into **milestones**:

- Sprint 1, Release 2.0, Q4 2025, etc.
 - Helps track progress toward a larger goal
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8. Track and Analyze

Use the **Insights** tab or third-party tools (e.g., ZenHub, Linear) to:

- Monitor issue trends
 - Track velocity and burndown
 - Analyze labels (e.g., # of bugs vs features)
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Summary Workflow

Activity	Tool/Feature
Create bug/feature	GitHub Issues + Labels
Track status	GitHub Projects board
Implement fix/feature	Pull Request (linked)
Group tasks	Milestones
Visualize work	Project + Insights

